

KADI SARVA VISHWAVIDYALAYA**B.E. SEMESTER-VIII (NEW) Examination JULY 2025****Subject Name: BIG DATA ANALYTICS****Subject Code: CE802-N****Date: 15/07/2025****Time: 10:30 AM to 01:30 PM****Total Marks: 70****Instructions:**

1. Answer each section in separate answer sheet.
2. Use of scientific calculator is permitted.
3. All questions are Compulsory.
4. Indicate clearly, the option you attempt along with its respective question number.
5. Use the last page of main supplementary of rough work.

Section-I

Q-1 (A) Explain in detail about the difference between Traditional Data Vs Big Data Business approach. [5]

(B) Explain types of Big Data in details. [5]

(C) What are the 5 V's of Big Data? Explain each in detail. [5]

OR

(C) Describe the various applications of Big Data Analytics in different major sectors. [5]

Q-2 (A) Explain the Bloom Filter with working in detail. [5]

(B) Explain the Flajolet Martin Algorithm. [5]

OR

Q-2 (A) Explain the Count-Distinct Problem. [5]

(B) List and explain different examples of Stream Data Sources. [5]

Q-3 (A) Explain Multihash Algorithm. [5]

(B) Define the Terms: 1. PIG 2. HIVE 3. MONGODB [5]

OR

Q-3 (A) Define the Terms: 1. Link spam 2. Hubs 3. Authorities [5]

(B) Draw and explain Hadoop Framework in detail. [5]

Section-II

- Q-4 (A) Discuss about Page Rank Algorithm in detail. [5]
(B) Explain Multistage Algorithm in detail. [5]
(C) Explain SimRank in detail. [5]

OR

- (C) Explain Social Network as Graph. [5]
Q-5 (A) What is NoSQL? Explain Characteristics of NoSQL database. [5]
(B) Differentiate between Master-slave Model Vs peer-to-peer Model. [5]

OR

- Q-5 (A) Explain share-nothing Architecture in Big Data. [5]
(B) Explain SQL Vs NOSQL. [5]

- Q-6 (A) Explain Matrix-Vector Multiplication by Map Reduce. [5]
(B) Explain Large scale file system organization. [5]

OR

- Q-6 (A) Explain Grouping and Aggregation by Map Reduce. [5]
(B) What is Big Data Visualization? Discuss the Big Data Visualization tools. [5]

******* All The Best *******